

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.* (BL3, BL4)

Activity Tool : Mock Interview (Low)
Assessment Tool Weightage: 30%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	5
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	5
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	5
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	5
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	5
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	5
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	5

8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	5
9	Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'	BL4 – Analyze	5
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	5

Code of Conduct:

I Sandra Sudevan(192424422) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sandrasudevan

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated

Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

1. Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan? — BL3

Answer:

Query processing converts an SQL query into an efficient execution plan.

Steps:

1. **Parsing and Translation** – Checks SQL syntax and converts the query into an internal representation.
2. **Query Optimization** – Generates different possible execution plans.
3. **Cost Estimation** – Estimates CPU, memory, disk I/O, and number of operations for each plan.
4. **Plan Selection** – Selects the plan with the lowest estimated cost.
5. **Query Execution** – The selected plan is executed to produce the result.

The optimizer considers factors such as **indexes, join methods, relation size, selectivity, and available memory** to choose the best plan.

2. Difference between heuristic optimization and cost-based optimization. When would you use each? — BL4

Heuristic Optimization

Uses predefined rules.

Does not calculate exact cost.

Cost-Based Optimization

Uses estimated execution cost.

Calculates cost for different plans.

Faster optimization.

Usually requires more optimization time.

Example: Perform selection before join.

Example: Choose hash join based on estimated cost.

May not always give the best plan.

Generally produces a more efficient plan.

Use: Heuristic optimization is useful when **quick optimization** is required. Cost-based optimization is preferred for **complex queries and large databases**, where performance is important.

3. How would you implement database connectivity in a real project? Explain JDBC and ODBC. — BL3

In a Java project, database connectivity can be implemented using **JDBC (Java Database Connectivity)**.

Steps:

1. Load the JDBC driver.
2. Create a database connection.
3. Create a PreparedStatement.
4. Execute SQL queries.
5. Process the ResultSet.
6. Close the connection and resources.

JDBC vs ODBC:

JDBC	ODBC
Designed for Java applications.	Language-independent database interface.
Uses Java APIs.	Uses ODBC APIs/drivers.
Platform independent with Java.	Depends more on ODBC driver configuration.
Commonly used in Java applications.	Commonly used by applications using ODBC-supported databases.

4. What are ACID properties? Give a real-world example where each is critical. — BL3

ACID properties ensure reliable database transactions.

1. **Atomicity** – A transaction is completed fully or not at all.
Example: In online banking, money should be debited and credited completely.
2. **Consistency** – A transaction must maintain valid database rules.
Example: A bank account should not have an invalid balance after a transaction.
3. **Isolation** – Concurrent transactions should not interfere incorrectly with each other.
Example: Two people should not be able to purchase the same last available seat.
4. **Durability** – Once a transaction is committed, its changes remain permanently.
Example: After a successful bank transfer, the transaction should remain recorded even after a system failure.

5. Explain dirty read, lost update, and unrepeatable read. How does 2PL solve them? — BL4

1. Dirty Read:

A transaction reads data that another transaction has modified but not yet committed.

2. Lost Update:

Two transactions update the same data, and one update overwrites the other.

3. Unrepeatable Read:

A transaction reads the same row twice and gets different values because another transaction modified it.

Two-Phase Locking (2PL) solves these problems by controlling access using locks.

It has two phases:

- **Growing Phase:** Transactions acquire locks but do not release them.
- **Shrinking Phase:** Transactions release locks but cannot acquire new locks.

2PL ensures **conflict serializability** and prevents incorrect concurrent updates.

6. What is deadlock in DBMS? Describe detection and prevention strategies. — BL4

A **deadlock** occurs when two or more transactions wait indefinitely for resources held by each other.

Example:

- Transaction T1 holds resource A and waits for B.
- Transaction T2 holds resource B and waits for A.

Deadlock Detection:

A **Wait-For Graph** is created:

- Nodes represent transactions.
- An edge $T1 \rightarrow T2$ means $T1$ is waiting for $T2$.
- If the graph contains a **cycle**, a deadlock exists.

Prevention strategies:

1. **Wait-Die:** Older transactions may wait, while younger transactions are rolled back.
2. **Wound-Wait:** Older transactions can force younger transactions to roll back.
3. **Resource Ordering:** Resources are requested in a predefined order to prevent circular waiting.

7. Compare Immediate Update and Deferred Update recovery. — BL4

Immediate Update	Deferred Update
Database may be updated before transaction commits.	Database is updated only after commit.
Requires UNDO and REDO during recovery.	Usually requires REDO only.
More flexible for updating data.	Simpler recovery process.
Can require more recovery processing.	Easier to maintain consistency.

Immediate Update: Preferred when transactions need to write changes before completion.

Deferred Update: Preferred when a **simple and reliable recovery mechanism** is desired and database changes can be postponed until commit.

8. Explain Shadow Paging. Advantages and limitations compared to log-based recovery. — BL3

Shadow Paging is a recovery technique that maintains two page tables:

- **Shadow page table** – Points to the original database pages.
- **Current page table** – Contains updated pages.

During a transaction, modified pages are written to new locations. The original pages remain unchanged. After successful commit, the system switches the pointer to the new page table.

Advantages:

- No need for a log.
- Recovery is fast.
- Simple rollback mechanism.

Limitations:

- Can cause fragmentation.
- Requires additional page management.
- Not efficient for very large databases.
- Maintaining page tables can create overhead.

Log-based recovery is generally more flexible and widely used in large database systems.

9. What is a serializability schedule? How do you test conflict serializability using a precedence graph? — BL4

A **serializable schedule** is a concurrent execution of transactions that produces the same result as some serial execution.

Conflict Serializability Test:

1. Create a node for every transaction.
2. Check pairs of conflicting operations.
3. If transaction T1 performs a conflicting operation before T2, draw an edge:
T1 → T2
4. Construct the **precedence graph**.
5. Check for cycles.

Result:

- **No cycle → Conflict serializable**
- **Cycle exists → Not conflict serializable**

Conflicting operations occur when two operations access the same data item and at least one operation is a write.

10. Explain Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why? — BL4

Protocol	Description
Basic 2PL	Locks are acquired during the growing phase and released during the shrinking phase.
Strict 2PL	All exclusive (X) locks are held until the transaction commits or aborts.
Rigorous 2PL	Both shared (S) and exclusive (X) locks are held until commit or abort.

Basic 2PL: Guarantees conflict serializability but may cause cascading rollbacks.

Strict 2PL: Prevents dirty reads and cascading rollbacks and simplifies recovery.

Rigorous 2PL: Provides even stronger control because all locks are retained until the transaction finishes.